

Maximum Sharpe Portfolio Weights

The *Sharpe* ratio is equal to the ratio of excess returns divided by the portfolio standard deviation:

$$SR = \frac{\mathbf{w}^T \boldsymbol{\mu}}{\sigma}$$

Where $\boldsymbol{\mu} = \bar{\mathbf{r}} - r_f$ is the vector of mean excess returns (in excess of the risk-free rate r_f), $\mathbf{w}$ is the vector of portfolio weights, and $\sigma = \sqrt{\mathbf{w}^T \mathbb{C} \mathbf{w}}$, where $\mathbb{C}$ is the covariance matrix of returns.

We can calculate the *maximum Sharpe* portfolio weights by setting the derivative of the *Sharpe* ratio with respect to the weights, to zero:

$$d_w SR = \frac{1}{\sigma} (\boldsymbol{\mu}^T - \frac{(\mathbf{w}^T \boldsymbol{\mu})(\mathbf{w}^T \mathbb{C})}{\sigma^2}) = 0$$

We then get:

$$(\mathbf{w}^T \mathbb{C} \mathbf{w}) \boldsymbol{\mu} = (\mathbf{w}^T \boldsymbol{\mu}) \mathbb{C} \mathbf{w}$$

We can multiply the above equation by $\mathbb{C}^{-1}$ to get:

$$\mathbf{w} = \frac{\mathbf{w}^T \mathbb{C} \mathbf{w}}{\mathbf{w}^T \boldsymbol{\mu}} \mathbb{C}^{-1} \boldsymbol{\mu}$$

We can finally rescale the weights so that they satisfy the linear constraint $\mathbf{w}^T \mathbf{1} = 1$:

$$\mathbf{w} = \frac{\mathbb{C}^{-1} \boldsymbol{\mu}}{\mathbf{1}^T \mathbb{C}^{-1} \boldsymbol{\mu}}$$

These are the weights of the *maximum Sharpe* portfolio, with the vector of mean excess returns equal to $\boldsymbol{\mu}$, and the covariance matrix equal to $\mathbb{C}$.

The *maximum Sharpe* portfolio is an *efficient portfolio*, and so its mean return is equal to some target return r_t : $\bar{\mathbf{r}}^T \mathbf{w} = \sum_{i=1}^n w_i r_i = r_t$.

The mean return of the *maximum Sharpe* portfolio is equal to:

$$r_t = \bar{\mathbf{r}}^T \mathbf{w} = \frac{\bar{\mathbf{r}}^T \mathbb{C}^{-1} \boldsymbol{\mu}}{\mathbf{1}^T \mathbb{C}^{-1} \boldsymbol{\mu}} = \frac{\bar{\mathbf{r}}^T \mathbb{C}^{-1} (\bar{\mathbf{r}} - r_f)}{\mathbf{1}^T \mathbb{C}^{-1} (\bar{\mathbf{r}} - r_f)} = \frac{\bar{\mathbf{r}}^T \mathbb{C}^{-1} \mathbf{1} r_f - \bar{\mathbf{r}}^T \mathbb{C}^{-1} \mathbf{1} \bar{\mathbf{r}}}{\mathbf{1}^T \mathbb{C}^{-1} \mathbf{1} r_f - \bar{\mathbf{r}}^T \mathbb{C}^{-1} \mathbf{1}} = \frac{c_{r1} r_f - c_{rr}}{c_{11} r_f - c_{r1}}$$

The above formula calculates the target return r_t from the risk-free rate r_f .

Optimal Portfolios Under Zero Correlation

If the correlations of returns are equal to zero, then the covariance matrix is diagonal:

$$\mathbb{C} = \begin{pmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{pmatrix}$$

Where σ_i^2 is the variance of returns of asset i .

The inverse of $\mathbb{C}$ is then simply:

$$\mathbb{C}^{-1} = \begin{pmatrix} \sigma_1^{-2} & 0 & \cdots & 0 \\ 0 & \sigma_2^{-2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^{-2} \end{pmatrix}$$

The *minimum variance* portfolio weights are proportional to the inverse of the individual variances:

$$w_i = \frac{1}{\sigma_i^2 \sum_{i=1}^n \sigma_i^{-2}}$$

The *maximum Sharpe* portfolio weights are proportional to the ratio of the excess returns divided by the individual variances:

$$w_i = \frac{\mu_i}{\sigma_i^2 \sum_{i=1}^n \mu_i \sigma_i^{-2}}$$

The portfolio weights are proportional to the *Kelly ratios* - the excess returns divided by the variances:

$$w_i \propto \frac{\mu_i}{\sigma_i^2}$$